Lesson 02: The Goals of Science; Science and Common Sense

English Stage 4 (Psychology) · Unit 1: Psychology as a Science

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 1.4 Goals of Science; 1.5 Science and Common Sense.

In this lesson

You will learn the three goals of scientific research (to describe, to predict and to explain), the difference between basic and applied research, and why common sense is not enough to understand behavior. On the language side, you will practice reporting research findings ("research has shown that …") and contrasting popular beliefs with evidence.

Learning goals

After this lesson you should be able to:

- Describe the three goals of science and give an example of each.
- Distinguish basic from applied research and explain why the distinction is not always clear-cut.
- Explain why folk psychology is often wrong, with reference to heuristics, confirmation bias and wishful thinking.
- Define skepticism and tolerance for uncertainty and apply them to a claim.
- Report findings with the present perfect and contrast beliefs with evidence (*although, however, instead, yet*).

Key language

Term or phrase	Meaning	Example
to describe	the first and most basic goal of science, achieved by making careful observations	A survey found which conditions medical marijuana patients treat.
to predict	to use a regular relationship between two behaviors or events to say whether one will occur	A medical marijuana user probably experiences pain.
to explain	the third and ultimate goal: determining the causes of behavior	Does marijuana reduce inflammation, which in turn reduces pain?
phenomenon (plural: phenomena)	a general result that has been observed reliably in empirical research	classical conditioning; obedience to authority

Term or phrase	Meaning	Example
basic research	research conducted mainly to achieve a more detailed and accurate understanding of behavior, not to solve a practical problem	Mehl's study of talkativeness
applied research	scientific research conducted primarily to solve some practical problem	research on cell phone use and driving
folk psychology	people's intuitive beliefs about human behavior and mental processes	"Letting anger out makes you feel better."
heuristics	mental shortcuts that we use in forming and maintaining beliefs	If experts agree and it makes sense, we assume it is true.
confirmation bias	the tendency to notice and remember evidence that is consistent with what we already believe and to ignore evidence that is inconsistent with it	noticing talkative women and silent men
skepticism	a critical-thinking attitude: considering alternatives and searching for evidence before accepting that a claim is true	Does an allowance teach responsibility, or only spending?
tolerance for uncertainty	a critical-thinking attitude: withholding judgment when there is insufficient evidence	There is no evidence either way about allowances.
research has shown that ...	introduces an established finding	Research has shown that false confessions are surprisingly common.
it turns out that ...	introduces a result, often a surprising one	It turns out that there is no evidence either way.
... is compounded by the fact that ...	a problem is made worse by a second fact	This is compounded by the fact that we focus on confirming cases.

Rules and explanations

The ideas

The three goals of science (1.4). The book's example is research on medical marijuana patients (Sexton, Cuttler, Finnell, & Mischley, 2016).

Goal	What it involves	Example
Describe	the first and most basic goal: careful observation (records, surveys)	Surveys of patients found that the primary symptom they treat is pain, followed by anxiety and depression.
Predict	once two behaviors or events are systematically related, using one to predict the other	A person who uses medical marijuana probably experiences pain. Predictions are not 100% accurate, but if the relationship is strong, they are better than chance.

Goal	What it involves	Example
Explain	the third and ultimate goal: determining causes and mechanisms	Does marijuana reduce inflammation, which in turn reduces pain, or does it only reduce the distress associated with pain?

Basic versus applied research. Basic research aims at understanding for its own sake (Mehl's study of talkativeness). Applied research addresses a practical problem: research on cell phone use while driving was prompted by safety concerns and has led to laws. The distinction is not always clear-cut (see Reading question 3).

Can we rely on common sense? (1.5). Our intuitive beliefs about behavior are called **folk psychology**. Much of it is probably reasonably accurate, but much of it is not:

- Most people believe anger is relieved by "letting it out" (punching something, screaming). Research has shown that this tends to leave people *more* angry (Bushman, 2002).
- Most people believe no one would confess to a crime they had not committed unless tortured. In fact, false confessions are surprisingly common (Kassin & Gudjonsson, 2004).
- Lilienfeld and colleagues (2010) list many myths, for example that we use only 10% of our brain power or that crimes increase during full moons.

Why are our intuitions so often wrong? (Gilovich, 1991). This is itself an empirical question, and research has identified several factors:

1. **Limited powers.** Accurate beliefs need more observation, memory and analysis than we naturally possess, so we rely on **heuristics**: if a belief is widely shared, endorsed by "experts" and intuitively sensible, we assume it is true.
2. **Confirmation bias.** We notice and remember cases that confirm our beliefs (talkative women, silent men) and ignore or forget the others.
3. **Wishful thinking.** We believe things partly because it would be nice if they *were* true. Calorie-reducing diets are not an effective long-term treatment for obesity (Mann et al., 2007), but the belief gives people hope, or lets them feel good about their own "self-control".

Two critical-thinking attitudes. Scientists know they are as susceptible as anyone else to intuitive but incorrect beliefs, so they cultivate:

- **Skepticism.** A magazine claims that a weekly allowance helps children develop financial responsibility. A skeptic asks: might it merely teach them to spend money, or make them more materialistic? Is the author a researcher? Is any evidence cited? What does the research literature (all the published research in the field) say?
- **Tolerance for uncertainty.** There is no scientific evidence that an allowance causes financial responsibility, nor that it causes materialism. Scientists accept that there is much they do not know.

Language focus

1. Reporting established findings. To report what research has established, the book uses the present perfect (*has shown, has found*) followed by a *that*-clause in the present simple, because the finding is still true now:

- "Scientific research, however, **has shown that** this approach **tends to** leave people feeling more angry, not less."
- "extensive empirical research **has shown that** false confessions **are** surprisingly common"
- "research involving surveys of medical marijuana patients **has been conducted and has found that** the primary symptom … **is** pain"

For one completed study, the book uses the past simple: "these researchers **conducted** a study in which female and male college students … **wore** audio recorders" (1.1).

2. Contrasting belief and evidence.

- "**Although** much of our folk psychology is probably reasonably accurate, it is clear that much of it is not."
- "many people believe that calorie-reducing diets are an effective long-term treatment for obesity, **yet** a thorough review … has shown that they are not"
- "Being skeptical does **not** mean being cynical or distrustful, **nor** does it mean questioning every belief … **Instead**, it means pausing to consider alternatives …"

> **Note for Arabic speakers** - Arabic has no separate present perfect. Use it for research whose findings still hold ("Research **has shown** that …"), and the past simple for one study at a stated time ("In 2007, Mehl and colleagues **found** …"). - After *nor* at the start of a clause, the auxiliary comes before the subject: "nor **does it** mean", not "nor it means".

Practice

1. Reading questions

1. Why will predictions about medical marijuana users not be 100% accurate, and when are they still useful?
2. Is Mehl's research basic or applied? Is research on cell phones and driving basic or applied? Give the book's reasons.
3. Why is the distinction between basic and applied research "not always clear-cut"? Give the book's two examples.
4. What do most people believe about anger and about false confessions, and what has research shown? (1.5)
5. Why do we rely on heuristics, according to 1.5?
6. How does confirmation bias keep the "talkative women" stereotype alive?
7. Why might people continue to believe in dieting despite the evidence?
8. What does skepticism *not* mean, and what does it mean?
9. Why is uncertainty "exciting from a scientific perspective"?

2. Book exercises. Sections 1.4 and 1.5 have no exercises of their own; Chapter 1's exercises are in Section 1.7 (Lesson 03).

3. Language exercise

A. Rewrite three of the "Great Myths" in 1.5 as reported findings, beginning "Scientific research has shown that ...".

B. Complete each sentence with *although, instead, nor* or *yet*.

1. Being skeptical does not mean being cynical, ____ does it mean questioning every belief. ____, it means pausing to consider alternatives.
2. ____ much of our folk psychology is probably reasonably accurate, much of it is not.
3. Many people believe diets are an effective long-term treatment for obesity, ____ a thorough review has shown that they are not.

C. Choose the better verb form and explain why.

1. In 2007, Mehl and colleagues (*found / have found*) that women and men speak about the same number of words per day.
2. Scientific research (*showed / has shown*) that venting anger tends to leave people more angry.

Speak with your tutor

- "Give me six research questions and let me say which goal of science each one serves."
- "PSYC 110 Lecture 10 (Evolution, Emotion, and Reason: Evolution and Rationality) describes heuristics and a confirmation-bias study about a child custody decision. Ask me to explain how confirmation bias works in that study and in the 'talkative women' example from Section 1.5."
- "Give me a claim from a popular magazine. I will respond with skepticism: alternatives and questions about evidence. Then tell me whether I was skeptical without being cynical."
- "Is research on sleep and memory in students basic or applied? Argue the opposite of whatever I say."

Writing task

Project warm-up. Write one paragraph (150 to 220 words) that prepares your research project (sleep and memory in university students, or your own topic) for Lesson 04.

Include:

- three research questions about your topic: one to describe, one to predict and one to explain (example: "How many hours do first-year students sleep on weeknights?"; "Do students who sleep fewer hours recall less of what they study?"; "Does sleep after studying cause students to remember more?");
- whether your interest is basic, applied or both, and why;
- one common belief about your topic, examined with skepticism: an alternative explanation, and what evidence you would look for;
- one sentence that shows tolerance for uncertainty ("I do not yet know whether ...").

If you work on your own topic, follow the same steps for it.

Self-editing checklist:

- Does each question clearly serve one goal (describe, predict or explain)?
- Is my "explain" question about a cause?
- Did I justify basic or applied?
- Did I offer at least one alternative explanation for the common belief?
- Did I use the present perfect only for findings that still hold, and the past simple for single past studies?
- Did I use at least one of *although, however, yet, instead* correctly?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- PSYC 110 Lecture 10, Evolution and Rationality (heuristics, availability bias and confirmation bias in Chapter 5, from 39:48): https://oyc.yale.edu/psychology/psyc-110/lecture-10

Answer key

1. Reading questions

1. Not every user experiences pain, so predictions from a relationship are never 100% accurate; but if the relationship between medical marijuana use and pain is strong, the predictions will have greater than chance accuracy (1.4, second goal).
2. Mehl's research is basic: it aims at a more accurate understanding of behavior, not at a practical problem. Cell phone research is applied: it was prompted by safety concerns and has led to laws (1.4).
3. Basic research on sex differences in talkativeness could eventually affect how marriage therapy is practiced, and applied research on cell phones and driving could produce new insights into basic processes of perception, attention and action (1.4).
4. Anger: people believe "letting it out" relieves it, but research shows it tends to leave people more angry (Bushman, 2002). Confessions: people believe no one confesses falsely unless tortured, but false confessions are surprisingly common and occur for a variety of reasons (Kassin & Gudjonsson, 2004) (1.5, first paragraph).
5. Forming detailed and accurate beliefs requires powers of observation, memory and analysis to an extent that we do not naturally possess (1.5, "How Could We Be So Wrong?").
6. Once we believe women are more talkative, we notice and remember talkative women and silent men but ignore or forget silent women and talkative men, so the belief seems to be confirmed (1.5).
7. It would be nice if it were true: the belief gives people hope of losing weight if they are obese, or makes them feel good about their own "self-control" if they are not (1.5).
8. It does not mean being cynical or distrustful, or questioning every belief (which would be impossible). It means pausing to consider alternatives and to search for evidence, especially systematic empirical evidence, when enough is at stake (1.5).
9. If we do not know the answer to an interesting and empirically testable question, science, and perhaps even you as a researcher, may be able to provide the answer (1.5, last paragraph).

2. Book exercises. See Lesson 03.

3. Language exercise

A. (example answers)

- Scientific research has shown that the belief that people use only 10% of their brain power is incorrect.
- Scientific research has shown that it is not true that students learn best when teaching styles are matched to their learning styles.
- Scientific research has shown that, contrary to popular belief, psychiatric admissions and crimes do not increase during full moons.

B. 1. nor; Instead. 2. Although. 3. yet.

C. 1. *found*: one study at a stated past time (2007). 2. *has shown*: the body of research, whose finding is still true now.

My notes